1. DATA WAREHOUSE (DW) - Concept & Overview

① What is a Data Warehouse ?

- A centralized, integrated, subject-oriented repository that stores current and historical data from multiple sources for analytics, reporting and decision making.
- Optimized for analytical queries (OLAP), not for day-to-day transactional processing (OLTP).
- Provides a single source of truth with consistent definitions.
- Mostly read-oriented for end users, but data is loaded/updated via controlled ETL/ELT pipelines (inserts, updates, merges).
- Can store transaction-level historical data (e.g., individual orders).

Key Points

- ✓ Historical + current data
- ✓ Integrated from multiple sources
- ✓ Read optimized (not OLTP)
- ✓ Single source of truth
- ✓ Supports complex queries
- ✓ Non volatile for users
- ✓ Loaded via ETL/ELT

② Why do we need a Data Warehouse ?

- Different source systems store data in different formats, levels and definitions.
- Need a consistent, integrated view of data across the organization.
- Enables trend analysis, forecasting, and better decision making.
- Separates analytical workload from operational systems.
- Supports enterprise-wide analytics and ML/AI use cases.

③ DW Architecture & Data Flow (Typical)

Source Systems (OLTP, APIs, Files, Applications)

- Databases
- APIs
- Files (CSV/JSON)
- Applications

→ **Ingestion ETL / ELT**

- Extract
- Validate
- Clean
- Transform
- Standardize
- Integrate
- Load (Batch/ Incremental/CDC)

→ **Data Warehouse** (Historical, Integrated, Cleaned)

→ **Consumption BI / Analytics**

- Reports
- Dashboards
- Ad-hoc Queries
- Data Marts
- ML / Advanced Analytics

④ Layers in a Modern DW (3-tier)

Top Layer (Presentation Layer)

- BI tools (Power BI, Tableau, Looker etc.)
- Dashboards
- Reports
- Ad-hoc queries

Middle Layer (Semantic / Analytical Layer)

- Business logic
- Aggregations
- Semantic models
- Cubes (in some systems)
- Query optimization

Bottom Layer (Data Warehouse Layer)

- RDBMS / Columnar store (e.g., Snowflake, BigQuery)
- Integrated historical data
- ETL/ELT pipelines
- Storage & compute

2. DATA WAREHOUSE - Characteristics, Types, Advantages

① Characteristics of a Data Warehouse (SITN)

Characteristic	Meaning (Interview Friendly)	Example
Subject-oriented	Organized around business subjects (Customer, Sales, Product etc.)	Sales, Finance, Customer
Integrated	Combines data from multiple sources with consistent formats, keys, data types and business definitions	Same customer across CRM and ERP is identified and standardized
Time-variant	Stores historical data (years) and keeps old data for trend analysis	Sales from 2020, 2021, 2022, 2023, 2024.
Non-volatile	Read-only for end users, data is loaded/updated via controlled pipelines, not by day-to-day transactions	Users cannot update past sales from a report

② Types of DW Architecture (1-tier, 2-tier, 3-tier)

1-Tier (Rarely used)

BI Tool → Data Warehouse (RDBMS)

- Minimal separation
- Least amount of data
- Not used or rarely used

2-Tier

BI Tool → Data Warehouse (RDBMS)

- BI layer directly interacts with DW
- Simple architecture
- Suitable for small to medium scale

3-Tier (Most widely used)

BI Tools (Reports/Dashboards) → Semantic / Analytical Layer (OLAP) → Data Warehouse (RDBMS / Columnar)

- Separation of storage, business logic and presentation
- Better scalability and performance
- Most commonly used in enterprises

③ Advantages of Data Warehouse

- ✓ Integrates multiple sources
- ✓ Supports historical analysis
- ✓ Single source of truth
- ✓ Consistent business metrics
- ✓ Separates analytical workload from OLTP
- ✓ Enables organization-wide analytics
- ✓ Supports advanced analytics / ML

④ Disadvantages of Data Warehouse

- ✗ High initial cost and complexity
- ✗ Requires data modeling and governance
- ✗ Storage and compute cost (large data)
- ✗ Data freshness depends on pipeline design
- ✗ Needs skilled resources
- ✗ Not suitable for high-frequency transactions

3. DATA MART - Concept, Types, Implementation

① What is a Data Mart ?

- A smaller, focused, subject-oriented subset of data from the enterprise data warehouse (or sometimes directly from source systems) designed for a specific department, business unit or subject (e.g., Sales, Finance, Marketing).
- Provides analytical capabilities to a specific group of users without the complexity of the entire DW.
- Faster to implement, easier to use and cheaper to maintain.

Key Takeaways

- ✓ Subset of DW (or source)
- ✓ Focused on a department
- ✓ Simpler and faster
- ✓ Can be built as tables/views
- ✓ Not mandatory for every org

② Types of Data Mart

Dependent Mart (from DW)

Enterprise DW → Sales Mart

- Data comes from central DW
- Most common in enterprises
- Consistent definitions
- Governance at DW/mart level

Independent Mart (from source)

Source Systems (OLTP / Files / APIs) → Sales Mart

- Built directly from sources
- Used in small orgs or specific teams
- Can lead to inconsistent definitions
- Counters the idea of DW

Hybrid Mart (from DW + source)

DW → Source; DW + Source → Sales Mart

- Combines data from DW and other sources
- Useful when some data is not available in DW
- More complex to maintain

③ Steps to Implement a Data Mart

1. Design	2. Define	3. Construct	4. Populate	5. Access	6. Manage
Gather business & technical requirements. Define scope and subject area.	Select subset of data. Design logical/physical model (tables, views). Define metrics & grain.	Create database objects (tables, views, indexes etc.)	Load data from DW or source systems. Set up metadata and access layer.	Provide querying capabilities & UI (BI tools, reports).	Monitor, maintain, refresh data. Handle changes and user access.

④ Advantages of Data Mart

- ✓ Focused and faster queries for a specific team
- ✓ Smaller data volume → cheaper to maintain
- ✓ Easier and quicker to implement
- ✓ Agile and adaptable to changing needs
- ✓ Reduces load on enterprise DW
- ✓ Enables department-specific security
- ✓ Empowers business users with self-service analytics

⑧ Disadvantages of Data Mart

- ✗ Limited subject/domain scope
- ✗ Cannot provide enterprise-wide view
- ✗ Risk of creating too many data marts (data junkyard)
- ✗ Potential inconsistencies if not governed properly
- ✗ Additional maintenance and pipelines
- ✗ Limited scope compared to enterprise DW.

4. DATA WAREHOUSE vs DATA MART & Common Concepts

① DW vs Data Mart (Comparison)

Aspect	Data Warehouse	Data Mart
Scope	Enterprise-wide	Specific department / subject
Subjects	Multiple business subjects	Narrower subject area
Data sources	Multiple operational systems	From DW, source systems or both
Size	Larger	Smaller subset
Purpose	Organization-wide analytics	Focused department analytics
Governance	Enterprise level	Should follow enterprise standards
Example	Customer + Orders + Payments	Sales Performance Mart

② Important Related Concepts

Enterprise Data Warehouse (EDW)

- Large, centralized DW across multiple domains.
- Consistent, governed data and metrics.
- Supports cross-department reporting and analytics.

ETL vs ELT

- ETL: Transform before loading into DW. Used in traditional systems.
- ELT: Load raw/semi-structured data first, then transform. (Snowflake, BigQuery, Databricks).

Historical vs Current Data

- DW can store both current and historical data.
- Retention depends on business requirements.
- Time-variant allows trend analysis.

Metadata

- Information about data (table definitions, column meaning, lineage, load timestamp, ownership).
- Helps in governance, data discovery and impact analysis.

Data Freshness / Latency

- Can be batch (daily), hourly or near real-time depending on business SLA.
- Depends on pipeline design and source availability.

Data Quality

- Validation, reconciliation, completeness, duplicates, correctness, consistency checks.
- Critical for trusted analytics.

Scalability & Performance

- Modern DWs use distributed processing, columnar storage, separation of compute and storage.
- Supports large volumes and complex queries.

5. SCENARIO BASED INTERVIEW QUESTIONS (with Answers)

Q1. Build a DW for an e-commerce company

Q: We have customer data in CRM, orders in MySQL and payments in another DB. How would you build a DW?

A: Understand sources, business keys and volume.
- Ingest via batch/incremental/CDC.
- Validate, clean and standardize data.
- Resolve customer identities across systems.
- Integrate into analytical models.
- Validate counts and business rules. Expose to BI tools / data marts.

Q2. Why not query OLTP directly?

Q: Why not run analytical queries directly on the production OLTP database?

A: OLTP is optimized for frequent transactions. Complex queries can impact performance.
- Lacks historical data and integration.
- DW provides workload isolation and consistent metrics.

Q5. Create a Sales Mart or not?

Q: Company has an EDW. Sales team needs faster reporting. Would you create a Sales Mart?

A: First check if issue is query design, compute capacity or missing aggregates.
- If team needs domain-specific queries, specialized aggregates or security isolation, a dependent mart is a good option.
- Don't create a mart without understanding the actual problem.

Q4. Revenue mismatch between teams

Q: Sales reports ₹10 Cr revenue, Finance reports ₹9.5 Cr. How would you investigate?

A: Check revenue definitions (gross/net, tax, discounts, refunds).
- Verify reporting period, time zone, currency.
- Check filters, transaction statuses.
- Reconcile source to target counts.
- Align business definitions and governed metrics.

Q5. Startup with only Marketing analytics

Q: A startup needs analytics for only Marketing. Would you build a full EDW?

A: Not necessarily.
- An independent marketing mart can be faster and cost-effective.
- Design with standard keys/metrics so it can later integrate with a broader DW.

Q6. Many inconsistent data marts

Q: Company has 20 independent marts with different customer/revenue numbers. What would you do?

A:
- Identify duplicated datasets and conflicting logic.
- Establish canonical definitions and governance.
- Introduce a central EDW or shared datasets.
- Migrate/align existing marts to trusted data.

Q7. Real-time or frequent refresh

Q: DW refreshes daily, but dashboard needs data every 15 minutes. What would you change?

A: Check which datasets need 15-min freshness.
- Use incremental ingestion or CDC.
- Implement more frequent transformations.
- Monitor end-to-end latency and SLA.

Q8. Incorrect historical data

Q: A transformation changed historical sales. How would you recover?

A: Identify impacted partitions/records.
- Trace transformation and source lineage.
- Correct logic and reprocess only affected period.
- Validate counts and financial totals.
- Refresh downstream marts/reports if needed.

Q9. External API + DW in a Finance Mart

Q: Finance mart needs warehouse data and exchange rates from external API. Which type?

A: - Hybrid data mart (from DW + source).
- Validate exchange rates, effective dates.
- Align currency and business definitions.
- Alternatively, load exchange rates into DW if needed by multiple teams.

Q10. Small data but slow queries

Q: A mart has only 10 million rows, but queries are slow. Why?

A: - Check query design and execution plan.
- Missing filters or aggregates.
- Joins and data model.
- Partitioning / clustering.
- Compute saturation or concurrency.
- Performance depends on more than size.